CASHWMaxSAT-DisjCom: Solver Description

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***Abstract*—This document describes the MaxSAT solver CASHWMaxSAT-DisjCom, submitted to the complete tracks (include unweighted and weighted track) of MaxSAT Evaluation 2024.**

I. Introduction

We developed an improved complete MaxSAT solver called CASHWMaxSAT-DisjCom based on the latest version UWr-MaxSat 1.5.4 [1], SCIP 8.1.0 [2] and CASHWMaxSAT-CorePuls 2023 [3]. In addition, CASHWMaxSAT-DisjCom used an unsatisfiable-core-based OLL procedure [4]–[7]. In this work, we propose two novel ideas to improve the last year's version of CASHWMaxSAT-CorePlus [3]. Based on CASHWMaxSAT-CorePlus, we optimize the parameter settings of SCIP, use an underlying SAT solver COMiniSatPS[1], and develop a new algorithm called CASHWMaxSAT-DisjCom.

- In [8], Extracting multiple disjoint cores in each iteration of the IHS algorithm can enhance the performance of solving MaxSAT. We have incorporated this idea into our solver. Initially, we obtain an unsatisfiable core using a SAT solver. Then, we remove this unsatisfiable core from the original assumptions and solve again using the SAT solver to get new disjoint unsatisfiable cores. This process is repeated until the used SAT solver returns SAT.
- We exclude some soft clauses that can be optimized by the Boolean multilevel optimization. For the remaining soft clauses, we calculate the standard deviation of their weight values. If the obtained standard deviation is less than the average value of their weight values, we add all the assumptions into the SAT solver.

II. Future work

First, we could use a simplified version of MaxSAT local search solvers to improve the satisfied solution. Second, we could try to design a novel selection way for selecting an unsatisfiable-core on weighted cases.

[1]https://github.com/marekpiotrow/cominisatps

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